Fernando Moreno-Pino, PhD

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Positions

- **Intelligent Systems Lab, University of Bristol** — Bristol, UK
  Assistant Professor in Machine Learning — *Dec. 2025 - Present*
  - **Research**: Deep learning, probabilistic machine learning, and quantitative finance.
  - **Academic leadership**: Deputy Programme Director for the MSc in Data Science; Senior Postgraduate Taught (PGT) Tutor.
- **Oxford-Man Institute, University of Oxford** — Oxford, UK
  Research Associate / Associate Member — *Dec. 2025 - Present*
  - **Research**: Develops machine-learning methods for sequential and time-series modelling, with applications to quantitative finance.
- **Department of Computer Science, University College London** — London, UK
  Part-Time Lecturer in Financial Computing — *Dec. 2025 - Present*
  - **Teaching**: Teaches and supervises students in the MSc programmes in Computational Finance, Financial Risk Management, Financial Technology, and Emerging Digital Technologies.
- **Oxford-Man Institute, University of Oxford** — Oxford, UK
  Postdoctoral Researcher — *Aug. 2023 - Nov. 2025*
  - **Research**: Led multiple first-author projects developing machine-learning methods for spatiotemporal and sequential modelling in quantitative finance, with publications in leading machine-learning and quantitative-finance venues.
- **Mathematical Institute, University of Oxford** — Oxford, UK
  Teaching Assistant — *Aug. 2023 - Nov. 2025*
  - **Teaching**: Taught deep learning on the University of Oxford's MSc in Mathematical and Computational Finance.
- **Universidad Carlos III de Madrid** — Madrid, Spain
  Research & Teaching Associate, Signal Processing and Learning Group — *Sep. 2018 - July 2023*
  - **Supervisor**: Prof. Dr. Antonio Artés Rodríguez.
  - **Research & teaching**: Conducted research and contributed to teaching in UC3M's Signal Processing and Learning Group.
- **Oxford-Man Institute, University of Oxford** — Oxford, UK
  Visiting Researcher, funded by MAN Group — *May 2022 - Oct. 2022*
  - **Supervisor**: Dr. Stefan Zohren.
  - **Research**: Investigated and developed neural methods for asset-volatility forecasting and fill-probability estimation in limit order books.
- **Universidad Carlos III de Madrid** — Madrid, Spain
  Research Assistant, Signal Processing and Learning Group — *Dec. 2017 - Sep. 2018*
  - **Supervisor**: Prof. Dr. Antonio Artés Rodríguez.
  - **Research**: Applied machine-learning methods to human activity recognition.
- **Universidad de Málaga** — Málaga, Spain
  Research Assistant, Department of Programming Languages and Computer Science — *Jan. 2016 - Sep. 2016*
  - **Supervisor**: Prof. Dr. Pedro Merino Gómez.
  - **Research**: Conducted research on software for low-latency communication systems at the MORSE Research Group.

Education

- **Universidad Carlos III de Madrid** — Madrid, Spain
  PhD in Probabilistic Machine Learning and Deep Learning (cum laude) — *Sep. 2018 – May. 2023*
  - **Dissertation**: "Deep Attentive Time Series Modelling for Quantitative Finance"
  - **Advisors**: Prof. Dr. Antonio Artés Rodríguez and Prof. Dr. Pablo Martínez Olmos.
  - **Research**: Research focused on probabilistic machine learning, signal-processing methods in deep learning, and Transformer-based time-series modelling and forecasting, with applications to quantitative finance. Earlier work included heterogeneous models for high-dimensional data.
- **Universidad Carlos III de Madrid** — Madrid, Spain
  M.Sc. in Telecommunications Engineering (Electrical Engineering) — *Sep. 2016 – Jul. 2018*
- **Universidad de Málaga** — Málaga, Spain
  B.Sc. in Telecommunications Engineering (Electrical Engineering) — *Sep. 2012 – Jul. 2016*
  - **Graduated with Honors**: Best academic record of the class.

Publications

- **Moreno-Pino, F.** (2026). Tail Starvation in Recursive Discovery: A Boundary for Validation and Re-entry. In Workshop on AI for Meta-Science, Advances in Neural Information Processing Systems 40 (NeurIPS), 2026.
- **Moreno-Pino, F.** (2026). Agreement Without Information: Spurious Directional Priors in Foundation Models. In International Conference on Artificial Intelligence in Finance (ICAIF), 2026. Paper selected for Oral Presentation.
- **Moreno-Pino, F.** (2026). Validation Protects Truth, Not Coverage: Tail Starvation in Scientific Proposal Priors. In Chen Institute Symposium for AI Advancing Science & Society (AIAS+), 2026.
- Huang, S., Kaushik, P., Chen, M., Pan, H., Chehab, O., **Moreno-Pino, F.**, & Simchowitz, M. (2026). Nano World Models: A Minimalist Implementation of Future Video Prediction . arXiv preprint arXiv:2605.23993.
- **Moreno-Pino, F.***, Arroyo, Á.*, Waldon, H.*, Dong, X., Cartea, Á. (2024). Rough Transformers: Lightweight and Continuous Time Series Modelling through Signature Patching. In Advances in Neural Information Processing Systems 38 (NeurIPS), 2024.
- **Moreno-Pino, F.**, Zohren, S. (2024). DeepVol: Volatility Forecasting from High-Frequency Data with Dilated Causal Convolutions. In Quantitative Finance, Taylor & Francis, 2024.
- **Moreno-Pino, F.***, Arroyo, Á.*, Waldon, H.*, Dong, X., Cartea, Á. (2024). Rough Transformers: Lightweight Continuous-Time Sequence Modelling with Path Signatures. In "Next Generation of Sequence Modelling Architectures" workshop, ICML 2024.
- **Moreno-Pino, F.***, Arroyo, Á.*, Waldon, H.*, Dong, X., Cartea, Á. (2024). Rough Transformers for Continuous and Efficient Time-Series Modelling. In "Learning from Time Series for Health" workshop, ICLR 2024.
- **Moreno-Pino, F.**, Olmos, P. M., & Artés-Rodríguez, A. (2023). Deep Autoregressive Models with Spectral Attention. In Pattern Recognition, Elsevier, 2023.
- Jiménez Rama, Ó., **Moreno-Pino, F.**, Ramírez, D., Olmos, P.M. (2023). Interpretable Spectral Variational AutoEncoder (ISVAE) for time series clustering. arXiv preprint arXiv:2310.11940.
- Arroyo, Á*., Cartea, Á., **Moreno-Pino, F.*** & Zohren, S. (2023). Deep Attentive Survival Analysis in Limit Order Books: Estimating Fill Probabilities with Convolutional-Transformers. Presented at Euro Working Group on Commodities and Financial Modelling (EWGCFM) 2023 & in Quantitative Finance, Taylor & Francis, 2023.

- Martínez-García, M.*, **Moreno-Pino, F.***, Olmos, P. M., & Artés-Rodríguez, A. (2023). Sleep Activity Recognition and Characterization from Multi-Source Passively Sensed Data. arXiv preprint arXiv:2301.10156.
- **Moreno-Pino, F.**, Martínez-García, M., Olmos, P. M., & Artés-Rodríguez, A. (2022). Heterogeneous Hidden Markov Models for Sleep Activity Recognition from Multi-Source Passively Sensed Data. Accepted at ML4H 2022, collocated with NeurIPS.
- **Moreno-Pino, F.**, Sükei, E., Olmos, P. M., & Artés-Rodríguez, A. (2022). PyHHMM: A Python Library for Heterogeneous Hidden Markov Models. arXiv preprint arXiv:2201.06968, submitted to the Journal of Machine Learning Research, Machine Learning Open Source Software section.
- Ríos-Muñoz, G. R., **Moreno-Pino, F.**, Soto, N., M. Olmos, P. , Artés-Rodríguez, A., Ferández-Avilés, F., & Arenal, A. (2020). Hidden Markov Models for Activity Detection in Atrial Fibrillation Electrograms. In 2020 Computing in Cardiology (pp. 1-4). IEEE.
- **Moreno-Pino, F.**, Porras-Segovia, A., López-Esteban, P., Artés, A., & Baca-García, E. (2019). Validation of Fitbit Charge 2 and Fitbit Alta HR against polysomnography for assessing sleep in adults with obstructive sleep apnea. Journal of Clinical Sleep Medicine, 15(11), 1645-1653.

Others

- **Moreno-Pino, F.**, Artés-Rodríguez, A. (2019). Human Activity Recognition in Psychiatric Patients through Heterogeneous Hidden Markov Models. Machine Learning Summer School (MLSS), Moscow, Russia (Poster).
- **Moreno-Pino, F.**, Artés-Rodríguez, A. (2018). Sleep Activity Recognition through Hidden Markov Models. Data Science Summer School (DS3), Paris, France (Poster).

Teaching

- **School of Engineering Mathematics, University of Bristol** — Bristol, UK
  Tutor at the Practice-Oriented Artificial Intelligence Centre for Doctoral Training (CDT) — *Oct. 2026 - Present*
- **Department of Computer Science, University College London** — London, UK
  Part-Time Lecturer in Financial Computing — *Mar. 2026 - Present*
  - **Computational Finance MSc**: 2026 – Present.
  - **Financial Risk Management MSc**: 2026 – Present.
  - **Financial Technology MSc**: 2026 – Present.
  - **Emerging Digital Technologies MSc**: 2026 – Present.
- **School of Engineering Mathematics, University of Bristol** — Bristol, UK
  Lecturer at the MSc in Artificial Intelligence — *Dec. 2025 - Present*
  - **Intro to AI and Data Analytics (Unit Director)**: 2026-Present.
  - **Visual Analytics**: 2026-Present.
- **SPAR Research Program** — Online
  Mentor on AI Safety — *Jan. 2026 - Present*
  - **Guarding Against Malicious Fine-Tuning**: 2026-Present.
- **Engineering Science, University of Oxford** — Oxford, UK
  Tutor at the Intelligent Earth Centre for Doctoral Training (CDT) — *Jan. 2025 - Mar. 2025*
  - **Deep Learning for Time Series Forecasting**: Hilary Term, 2025.
- **Magdalen College, University of Oxford** — Oxford, UK
  Tutor at the "Oxford Study Abroad Programme" for Visiting BSc and MSc Students — *Sep. 2023 – Nov. 2025*
  - **Deep Learning**: 2023 – 2025.
- **Mathematical Institute, University of Oxford** — Oxford, UK
  Teaching Assistant at the MSc in Mathematical and Computational Finance — *Sep. 2023 – Nov. 2025*

*Denotes co-first authors with equal contributions.

- **Deep Learning** : 2023 – 2025.

- **BBVA** Madrid, Spain
  External Teaching Staff, Fundación Universidad Carlos III *Sep. 2021 – Present*
  - **Deep Generative AI Course**: 2025 – Present.
  - **Advanced Machine Learning and Feature Engineering Course**: 2022 – Present.
  - **Intermediate Machine Learning and Feature Engineering Course**: 2022 – Present.
  - **Natural Language Processing (NLP) Course**: 2021 – Present.

- **Universidad Carlos III de Madrid** Madrid, Spain
  Lecturer and Teaching Assistant, B.Sc. in Electrical Engineering *Sep. 2018 – July 2023*
  - **Signals and Systems**: 1st Semester 2023 (30 hours).
  - **Machine Learning II**: 1st Semester 2021 (38 hours).
  - **Bayesian Machine Learning, Modern Theory of Detection and Estimation**: 1st Semester 2018 – 2019 (61 hours).
  - **Communications Theory**: 1st Semester 2018 – 2019, 2023 (102 hours).
  - **Linear Systems**: 1st Semester 2018 (11 hours).

Honours, Awards, and Grants

- **Cosmos Institute Grants: Fast funding for AI prototypes (2026)**: Awarded $6,000 under the "AI for Truth-Seeking" track.
- **SPAR Research Program for AI Risks (2026)**: Awarded $2,000 to support the "Guarding Against Malicious Fine-Tuning: Hardening Models and Detecting Tampering" project.
- **MAN Group & OMI's Visitors Programme Fellowship (2022)**: Research visit to the Oxford-Man Institute during my PhD, funded by the OMI Visitors Programme (£7,200).
- **UC3M Research Mobility Fellowship (July-October 2022)**: Research visit to the University of Oxford (€4,716).
- **FPU Research Mobility Fellowship (May-July 2022)**: Research visit to the University of Oxford (€4,095).
- **FPU Grant (2018-2022)**: Doctoral funding from the Spanish Ministry of Education, awarded through a nationwide competitive selection process (€60,996.12).
- **Research Fellowship, Signal Processing and Learning Group, Universidad Carlos III de Madrid (2017)**: Research funding for an M.Sc. dissertation (€5,250).
- **Talentia Scholarship by "Junta de Andalucía" (2017, declined)**: Two-year scholarship offer for a Master of Engineering at Cornell University's New York campus (€62,092.68).
- **"Premios Extraordinarios de Fin de Estudios" (2016)**: Awarded by Universidad de Málaga for the highest academic record in the bachelor's cohort.
- **'Premios Ingenio', Finalist (2016)**: Finalist for the Andalusian award for best telecommunications engineering thesis.
- **Research Fellowship, Department of Computer Science and Programming Languages, Universidad de Málaga (2016)**: Research funding for a B.Sc. dissertation (€2,074).

Committees / Memberships

- **The London Initiative for Safe AI**: Since 2026 (Resident Member).
- **Andaluz.IA**: Since 2026 (Scientific Committee).
- **KE (Knowledge Exchange) Hub's Early Career Mathematical Scientists (ECMS) Forum**: Since 2026.
- **Effective Altruism Bristol: Mentor on AI Safety**: Since 2026.
- **AAAI Conference on Artificial Intelligence**: Since 2026 (Program Committee).
- **Chen Institute Symposium for AI Advancing Science and Society**: Since 2026.
- **Engineering and Physical Sciences Research Council (EPSRC)**: Since 2026 (Member of Peer Review College).
- **European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECMLPKDD)**: Since 2026 (Program Committee).
- **International Conference on AI in Finance (ICAIF)**: 2023, 2025 (Program Committee).

Reviewing

- **AAAI Conference on Artificial Intelligence**: Since 2026.
- **Chen Institute Symposium for AI Advancing Science and Society**: Since 2026.
- **The European Conference on Computer Vision**: Since 2026.
- **European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECMLPKDD)**: Since 2026.
- **Neural Networks**: Since 2025.
- **Neural Information Processing Systems (NeurIPS)**: Since 2024 (Reviewer and workshop proposals reviewer).
- **International Conference on Machine Learning (ICML)**: Since 2024.
- **International Conference on Learning Representations (ICLR)**: Since 2024.
- **Journal of Applied Mathematical Finance**: Since 2024.
- **Artificial Intelligence and Statistics (AISTATS)**: Since 2023.
- **ACM International Conference on AI in Finance (ICAIF)**: Since 2023.
- **Journal of Quantitative Finance**: Since 2023.
- **Pattern Recognition**: Since 2022.
- **AAAI Conference on Artificial Intelligence**: Since 2022.
- **IEEE Transactions on Neural Networks and Learning Systems**: Since 2021.
- **Journal of Biomedical and Health Informatics (JBHI)**: 2020-2023.

Talks

- **Practice-Oriented Artificial Intelligence Seminar, University of Bristol** — Bristol, UK
  Deep Attentive Sequential Modelling: Methods, Theory, and Applications — *Mar. 2026*
- **SIAM Conference on Financial Mathematics and Engineering** — Miami, US
  Generative AI in Finance — *Jul. 2025*
- **Department of Engineering Mathematics, University of Bristol** — Bristol, UK
  Deep Attentive Time Series Modelling for Quantitative Finance — *Jun. 2025*
- **Quantitative Risk Management and Mathematical Finance, University of Vienna** — Vienna, Austria
  Neural Dynamical Systems for Time-Series Data — *Apr. 2025*
- **Invited Talk at Citi Bank** — New York, US
  Rough Transformers: Lightweight and Continuous Time Series Modelling through Signature Patching — *Mar. 2025*
- **The Alan Turing Institute & Mathematical Institute, University of Oxford** — London, UK
  Bridging Rough Paths and Deep Learning: New Frontiers [Video] — *Nov. 2024*
- **Oxford-Man Institute, University of Oxford** — Oxford, UK
  Deep Autoregressive Models with Spectral Attention — *Mar. 2024*
- **Signal Processing and Learning Group, Universidad Carlos III de Madrid** — Madrid, Spain
  Deep Attentive Time Series Modelling for Quantitative Finance — *Apr. 2023*
- **Oxford-Man Institute, University of Oxford** — Oxford, UK
  DeepVol: Volatility Forecasting from High-Frequency Data with Dilated Causal Convolution — *Oct. 2022*
- **Signal Processing and Learning Group, Universidad Carlos III de Madrid** — Madrid, Spain
  Deep Autoregressive Models with Spectral Attention — *Oct. 2021*

Conferences, Summer Schools, and Others

- **SIAM** — Miami, US
  Conference on Financial Mathematics and Engineering — *Jul. 2025*
- **NeurIPS** — Vancouver, Canada
  Conference on Neural Information Processing Systems — *Dec. 2024*
- **ICML** — Vienna, Austria
  International Conference on Machine Learning — *Jul. 2024*
- **ICLR** — Vienna, Austria
  International Conference on Learning Representations — *May. 2024*
- **AISTATS** — Valencia, Spain
  Artificial Intelligence and Statistics — *Apr. 2023*
- **AI for Global Goals - University of Oxford** — Oxford, United Kingdom
  ML x Finance — *Aug. 2022*
- **University of Sheffield** — Sheffield, United Kingdom [Online]
  The Gaussian Process Summer School — *Sep. 2021*
- **University of Sheffield** — Sheffield, United Kingdom [Online]
  The Gaussian Process Summer School — *Sep. 2020*
- **Liège Université** — Liége, Belgium [Online]
  Machine Learning Frontiers in Precision Medicine (MLFPM) — *Sep. 2020*
- **ETH Zürich** — Basel, Switzerland
  Machine Learning Frontiers in Precision Medicine (MLFPM) — *Sep. 2019*
- **Skoltech** — Moscow, Russia
  Machine Learning Summer School (MLSS) — *Aug. 2019 – Sep. 2019*
- **École Polytechnique** — Paris, France
  Data Science Summer School (DS3) — *Jun. 2018*

Courses

- **University of California, Santa Cruz** — Coursera [Online]
  Bayesian Statistics: From Concept to Data Analysis, 4-week course — *Jul. 2021*
- **DeepLearning.AI** — Coursera [Online]
  Structuring Machine Learning Projects, 3-week course — *May 2018*
- **DeepLearning.AI** — Coursera [Online]
  Improving DNNs: Hyperparameter Tuning, Regularization and Optimization, 2-week course — *May 2018*
- **Universidad Internacional Menéndez Pelayo** — Barcelona, Spain
  English Immersion Program — *Apr. 2018*
- **DeepLearning.AI** — Coursera [Online]
  Neural Networks and Deep Learning, 4-week course — *Mar. 2018*
- **Stanford University** — Coursera [Online]
  Machine Learning, 11-week course — *Feb. 2018*
- **University of Washington** — Coursera [Online]
  Machine Learning: Classification, 7-week course — *Nov. 2017*
- **University of Washington** — Coursera [Online]
  Machine Learning: Regression, 6-week course — *Oct. 2017*
- **University of Washington** — Coursera [Online]
  Machine Learning Foundations, 6-week course — *Jul. 2017*
- **Nvidia Corporation, CUDA Fellows Program & Universidad de Málaga** — Málaga, Spain
  Technical Training Course: Parallel Programming of the GPU with CUDA — *Jul. 2016 – Aug. 2016*

Languages

- **Spanish**: Native language.
- **English**: Advanced, TOEFL: 102/120.
- **French**: Basic.

Projects

- **Heterogeneous Hidden Markov Model**: Python implementation of an HMM capable of handling heterogeneous and missing data: **https://github.com/fmorenopino/HeterogeneousHMM**, **https://pyhhmm.readthedocs.io/**.
- **VoIP calls**: C implementation of a Voice over IP call service for point-to-point audio conferencing. RTP over UDP was used: **https://github.com/fmorenopino/c_calls**.

Programming Skills

- **Languages**: Python, Matlab, C, C++ **Technologies**: Pytorch, Keras, scikit-learn, Jupyter, Git, LaTeX

Referees

- Dr. Álvaro Cartea, University of Oxford, UK.
- Dr. Stefan Zohren, University of Oxford, UK.
- Dr. Antonio Artés Rodríguez, Universidad Carlos III de Madrid, Spain.
- Dr. Pablo Martínez Olmos, Universidad Carlos III de Madrid, Spain.